Unlock the Lab: evaluation guide

How the workshop works, the six criteria and a glossary

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How the workshop works

‘Chocolate makes you smarter.’ Headlines promise miracles and warn of disasters, but how good is the research behind them?

In this workshop you read short descriptions of studies and judge their quality with six criteria that scientists use: Title, Access, Source, Theory, Methods & Data and Conclusion.

The studies are fictional. They were written for this workshop and are modelled on the strengths and flaws of real research, but none of them describes a real study.

Your two tasks

For each study you give two answers on a scale from 1 (very low quality) to 7 (very high quality):

1. **Predict the crowd.** Guess the average rating that workshop participants give the study. This is the answer that earns points; how it is scored is explained below.
2. **Give your own rating.** Judge the study yourself with the six criteria. Your own rating does not change your score, but it becomes part of the crowd that later participants predict.

After each study you see your score, the target with how it was worked out and the expert’s feedback on each of the six criteria.

How scoring works

Your prediction is scored against a target rating. The target combines the ratings that other participants have given the study so far with an expert’s rating, which counts as if 100 people had given it. Your own rating is never part of your target.

Because the expert counts so heavily, the target stays close to the expert’s rating until many people have taken part. Even with 20 ratings from other participants, the expert’s rating makes up 83% of the target.

A prediction that matches the target exactly scores 100 points. You lose 12 points for each point you are away from the target, so a prediction 1 point away scores 88 and one 2 points away scores 76.

As formulas, where p is your prediction, e the expert’s rating, n the number of other participants who have rated the study and S the sum of their ratings:

$$target = (S + 100 \times e) / (n + 100)$$
$$score = \max(0, 100 - \text{round}(12 \times |p - target|))$$

For example, if the expert rated a study 5 and 20 other participants have given it an average of 3 (a sum of 60), the target is (60 + 100 × 5) / (20 + 100) = 4.67. A prediction of 4 is 0.67 points away, so it scores 100 − round(12 × 0.67) = 92.

Leaderboard and finishing

Your results appear on a public leaderboard under a random nickname, such as ‘Curious Otter 42’. The leaderboard ranks people by their average score per study, once they have rated at least 10 studies.

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There are 48 studies. You can stop and see your results at any point after rating 10.

What matters most to you? You have 20 tokens to share between the six criteria you used. Give more tokens to the criteria that matter most to you when you judge a study.

The six criteria

- **Title:** Does the title describe the finding accurately, without hype?
- **Access:** Can anyone read the full study free of charge?
- **Source:** Was it published by a trustworthy, independent outlet?
- **Theory:** Does it build on earlier research to fill a real gap in knowledge?
- **Methods & Data:** Are the design, sample and controls adequate, and are the data open?
- **Conclusion:** Do the claims stay within what the data show?

Quality rubric

Criterion	Low (1–2)	Medium (3–5)	High (6–7)
Title	**Clickbait:** Uses words like 'Miracle', 'Proven', 'Cure'. Scarier or better than the actual data.	**Modest:** Describes the finding but might leave out limitations to sound more interesting.	**Accurate:** Boring but true. Describes exactly what happened (e.g. 'Correlation observed in mice').
Access	**Impossible:** Behind a paywall (£30+), no link provided or 'Contact author' with no reply.	**Difficult:** Requires university login, searching through specific databases or requesting via forums.	**Open:** One-click access to the full PDF. Free for everyone (Open Access).
Source	**Suspicious:** Marketing blogs, 'Predatory' journals (pay-to-publish) or corporate white papers.	**Unverified:** Preprints (not yet reviewed), or mid-tier journals that sometimes favour hype.	**Trusted:** Top-tier peer-reviewed journals, reputable independent research institutes or government bodies.
Theory	**Nonsense:** Ignores all previous science. Invents new laws of physics/biology without proof.	**Standard:** Repeats what we already know without adding much new value.	**Robust:** Fills a clear gap in knowledge. Uses past research responsibly to build a new argument.
Methods & Data	**Flawed & Closed:** Tiny sample size. No control group. Data is secret/hidden.	**Acceptable:** Decent sample size. Standard methods. Data available upon request.	**Rigorous & Open:** Preregistration, large sample size, gold-standard controls and fully available materials and data.
Conclusion	**Overblown:** Claims a fact based on a guess. Confuses correlation with causation.	**Logical:** Conclusion mostly fits the results but ignores some alternative explanations.	**Nuanced:** Very careful. Admits what they don't know. Claims only what the data proves.

From six criteria to one rating

Use these six criteria to judge each study, then weigh them up to reach your overall rating from 1 to 7. You can give some criteria more weight than others.

The three columns of the rubric use the same scale as your rating: Low (1–2), Medium (3–5) and High (6–7).

When you rate a study, the app asks: 'How good is this study overall? Weigh up the six criteria.'

After each study, the expert's feedback gives a band for each criterion and a summary of how the expert weighed them up, so you can compare your judgement with the expert's.

At the end you share 20 tokens between the six criteria to show which matter most to you.

Glossary

Terms you will meet in the studies.

Abstract

A brief summary at the beginning of a research paper outlining the study's purpose, methods, results and conclusions.

Actigraphy

Measuring sleep and activity with a motion sensor worn on the wrist. It is more objective than asking people how well they slept.

Blinding

Keeping people in a study unaware of who receives which treatment, so that expectations cannot sway the results. In a single-blind study the participants do not know; in a double-blind study neither the participants nor the researchers assessing them know.

Cherry-picking

Presenting only the results or studies that support a claim while leaving out those that do not.

Citation

A reference to another study or source, showing where information came from and allowing readers to verify claims.

Cohort Study

An observational study that follows a group of people over time, sometimes for decades, to see how their habits or circumstances relate to later outcomes such as illness.

Conflict of Interest

When researchers have financial, professional or personal interests that could influence their work. Good studies declare these transparently.

Confounding Variable

A third factor that influences both the supposed cause and the outcome, creating a link between them that is not causal. For example, wealth may explain why people who buy organic food are healthier.

Control Group

A comparison group that doesn't receive the treatment being tested, used to determine if an intervention actually works.

Correlation vs Causation

Correlation means two things happen together; causation means one causes the other. Correlation does NOT prove causation.

Cross-sectional Study

An observational study that measures people once, at a single point in time. It can show that two things go together but not which came first.

Data

The information collected during a study, which can include measurements, observations, survey responses or other recorded facts.

Effect Size

How large a difference or relationship is, not just whether there is one. For the common measure Cohen's *d*, about 0.2 counts as small, 0.5 as medium and 0.8 as large.

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Heterogeneity

How much the results of the studies in a review or meta-analysis differ from one another. When it is high, a single pooled figure should be read with caution.

Hypothesis

A testable prediction about what will happen in a study, based on existing knowledge or theory.

Impact Factor

The average number of times a journal's recent articles are cited by other research. It is a rough sign of a journal's prestige but says little about the quality of any single study.

Meta-analysis

A statistical method that pools the results of several studies on the same question into one overall estimate. It is often part of a systematic review.

Methods

The section describing how the study was conducted, including participants, materials, procedures and analysis techniques.

Observational Study

A study in which researchers record what people already do or experience, without assigning them to groups. It can reveal associations but cannot by itself show cause and effect.

Open Access

Research papers that are freely available to read online without a paywall or subscription. Anyone can access the full text immediately.

Open Data

When researchers make their raw data publicly available so others can verify results and conduct their own analyses.

Open Science Framework (OSF)

A free online platform where researchers can preregister studies and share their data, code and materials so that others can check and reuse them.

P-hacking

Trying many different analyses and reporting only those that reach statistical significance. It makes chance findings look real; preregistration helps to prevent it.

Paywall

A barrier that requires payment (often £30+) or institutional access to read a research paper. This limits who can verify the science.

Peer Review

A process where other experts in the field evaluate a research paper before publication to check its quality and validity.

Placebo

A dummy treatment, such as a sugar pill or a sham device, that looks like the real one but has no active ingredient. Comparing a treatment with a placebo shows whether it works beyond people's expectations (the placebo effect).

Predatory Journal

A publication that charges authors fees but skips proper peer review. Many reputable open-access journals also charge publication fees; the warning sign is the missing quality control. These journals prioritise profit over scientific quality.

Preprint

An early version of a research paper shared publicly before peer review. Useful for rapid communication but not yet vetted by experts.

Preregistration

Declaring your research plan and analysis methods publicly before collecting data, which reduces bias and questionable research practices.

Randomised Controlled Trial (RCT)

An experiment in which chance decides whether each participant receives the treatment being tested or joins a control group. Because the groups start out alike, differences in outcome can be attributed to the treatment.

Replication

Repeating a study to see if the same results occur. Replicable findings are more trustworthy than one-off results.

Results

The section presenting what the study found, including statistical outcomes and data visualisations.

Reverse Causality

When the supposed effect is actually the cause. For example, people who sleep badly may use their phones more because they are awake, rather than phone use causing poor sleep.

Sample Size

The number of participants (*N*) or observations in a study. Larger samples generally provide more reliable results.

Statistical Significance

A finding is called statistically significant when results at least this strong would rarely occur by chance if there were really no effect (by convention, $p < 0.05$). It does not show how large or important an effect is, and a non-significant result does not prove that there is no effect.

Systematic Review

A review that searches for all the studies on a question using a method set out in advance, assesses their quality and summarises the evidence. Unlike a narrative review, it is designed to avoid cherry-picking.

Theory

In science, a theory is a well-tested explanation of how something works, supported by evidence from many studies; in everyday speech the word often just means a guess. The rubric's Theory criterion asks something different: how well a study builds on existing research and whether it addresses a real gap in knowledge.

Variable

Any factor that can change or be measured in a study, such as age, temperature or test scores.

White Paper

A report published directly by a company, industry body or campaign group, usually without independent peer review. White papers are often used for marketing or lobbying.

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